



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - Jan - June 2024

UQ22CA723BB - Generative AI

Total Marks : 50.0

1.a. Compare various Generative AI models emphasising the application and the data type it operates on. (4.0 Marks)

1.b. What is A/B testing? How generative AI is involved in conducting A/B tests? Give an example. (5.0 Marks)

1.c. Write the steps of Generative AI in Data science Life cycle. (4.0 Marks)

2.a. Describe the functionality and applications of Adobe Firefly in the context of AI image generation. (4.0 Marks)

2.b. Explain the key capabilities of common models/tools for text generation. (4.0 Marks)

2.c. You are an amateur musician who has been struggling to produce a high-quality soundtrack for your new YouTube video, based on the following constraints.

1. You want the soundtrack to have a modern electronic feel with a blend of ambient sounds and upbeat rhythms.
2. You also need to remove some background noise from your vocal recordings and ensure the final mix sounds professional.

Outline the steps you would take to achieve this using the tools that you have studied. (4.0 Marks)

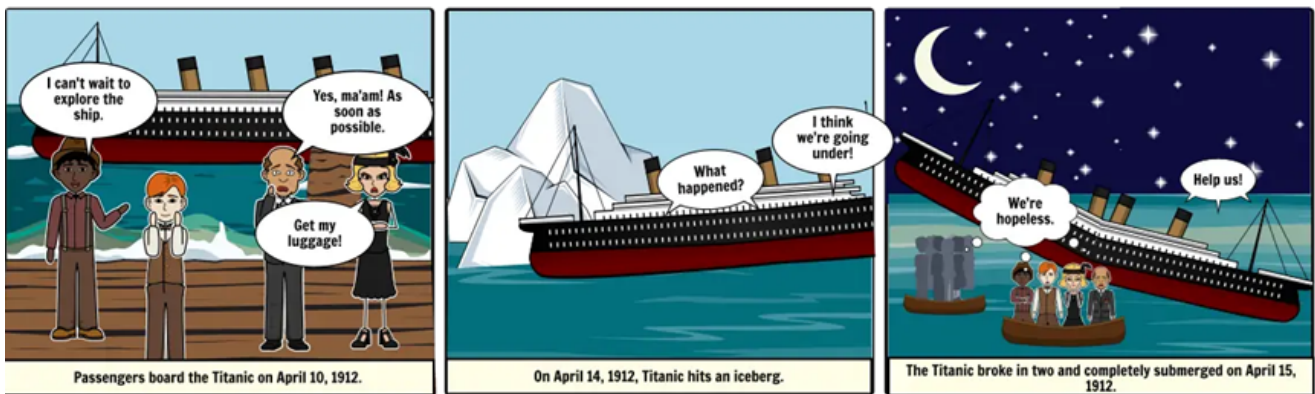
3.a. Imagine you are working for a customer support chatbot company. Your task is to design a prompt that helps the AI assist users in troubleshooting common issues with their internet connection. The prompt should guide the AI to ask relevant questions and provide step-by-step solutions. Create prompt sequences and explain how it will help the AI in providing effective customer support. (4.0 Marks)

3.b. Compare and contrast the approach of prompt design in IBM Watsonx.ai prompt lab and Scale AI's Spellbook. (5.0 Marks)

3.c. Describe the best practices of prompt creation. (4.0 Marks)

4.a. You are an AI prompt engineer tasked with designing prompts for an educational chatbot aimed at teaching elementary students basic arithmetic. The goal is to help the chatbot understand and apply the "Chain-of-Thought" approach to solve multi-step math problems more effectively. Walkthrough the various levels to design it. (6.0 Marks)

4.b.



This is sequence of images need to be developed. What sequence of text prompts you have to make to generate this story? (6.0 Marks)